

LAPORAN TUGAS DATA WAREHOUS JOBSHEET 3– Database Analytical



Disusun Oleh:

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2G

**D4 SISTEM INFORMASI BISNIS
TEKNOLOGI INFORMAS
POLITEKNIK NEGERI MALANG
2025**

A. Dimensi Waktu

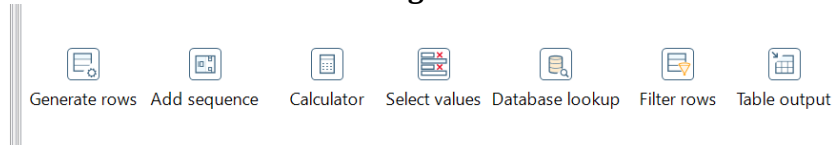
Praktikum

Proses bisnis pada LegendVehicle adalah 5 tahun. Sehingga data pada tabel dimdate yang harus tersedia adalah tanggal selama 5 tahun. Mulai dari 1 Januari 2023

1. Membuat database



2. Membuat table dimensi dimDate dengan membuat new Transformation



3. Konfigurasi Generate Rows

Step name: Generate rows

Limit: 1825

Never stop generating rows: ☐

Interval in ms (delay): 5000

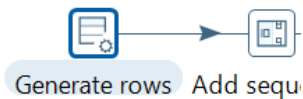
Current row time field name: now

Previous row time field name: FiveSecondsAgo

Fields :

#	Name	Type	Format	Length	Precision	Currency	Decimal	Group	Value	Set emp
1	CurrentData	Date	dd-MM-yyyy						01-01-2023	N

Buttons: Help, OK, Preview, Cancel



4. Add sequence

Step name: Add sequence

Name of value: incrementDay

Use a database to generate the sequence:

Use DB to get sequence? ☐

Connection: conn_dw_destination

Schema name: Schemas...

Sequence name: SEQ_ Sequences...

Use a transformation counter to generate the sequence:

Use counter to calculate sequence? ☒

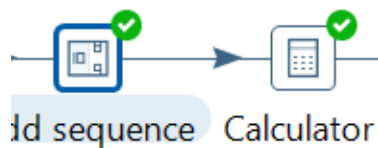
Counter name (optional):

Start at value: 0

Increment by: 1

Maximum value: 999999999

Buttons: Help, OK, Cancel



5. Calculator

Calculator

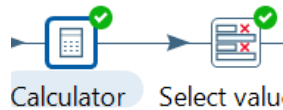
Step name: Calculator

☐ Throw an error on non existing files

Fields:

#	New field	Calculation	Field A	Field B	Field C	Value type	Length	Precision	Remove	Conversion mask	Decimal symbol	Grouping symbol
1	streamDate	Date A + 8 Days	CurrentDate	incrementDay		None			N			
2	streamYear	Year of date A	streamDate			None			N			
3	streamMonth	Month of date A	streamDate			None			N			
4	streamDay	Day of month of date A	streamDate			None			N			

Help OK Cancel



6. Select Values

Select values

Step name: Select values

Select & Alter Remove Meta-data

Fields :

#	Fieldname	Rename to	Length	Precision
1	CurrentDate			
2	incrementDay			
3	streamDate			
4	streamYear			
5	streamMonth			
6	streamDay			

Get fields to select Edit Mapping

Include unspecified fields, ☐

Help OK Cancel

Select values

Step name: Select values

Select & Alter Remove Meta-data

Fields to remove :

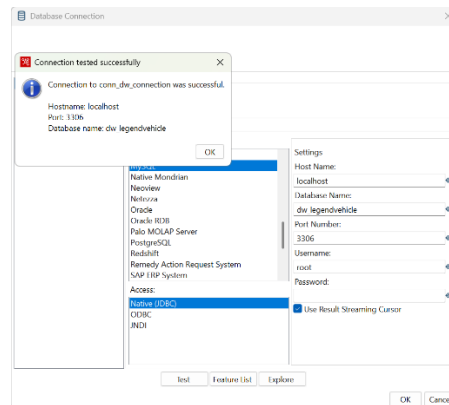
#	Fieldname
1	CurrentDate
2	incrementDay

Get fields to remove

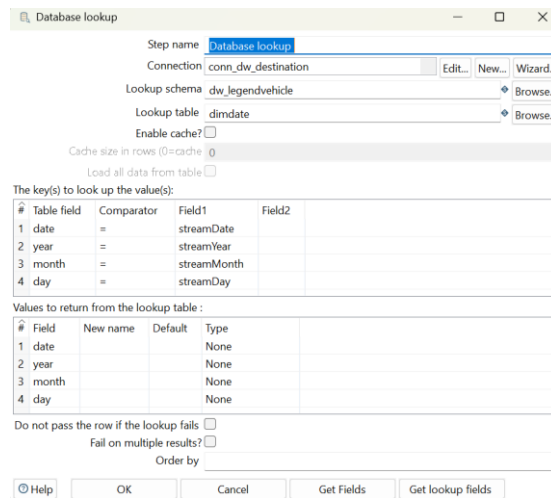
Help OK Cancel



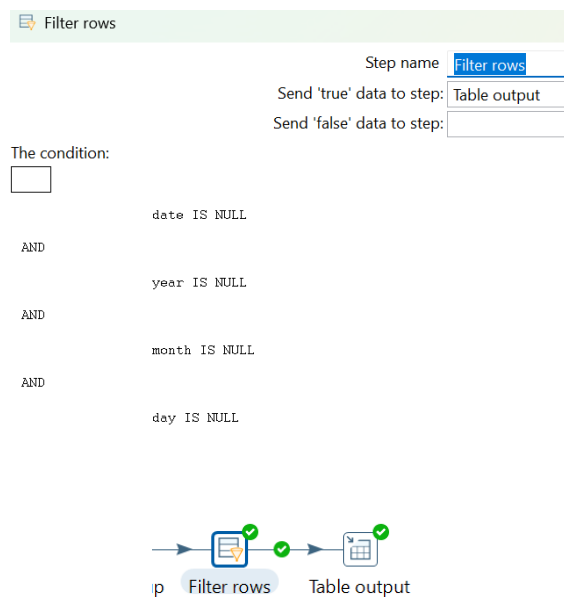
7. Connection Database



8. Database lookup



9. Filter Rows



10. Table output

Table output

Step name: **Table output**

Connection: **conn_dw_destination** [Edit...] [New...] [Wizard...]

Target schema: **dw_legendvehicle** [Browse...]

Target table: **dimdate** [Browse...]

Commit size: **1000**

Truncate table: ☐

Ignore insert errors: ☐

Specify database fields: ☒

Main options | Database fields

Fields to insert:

#	Table field	Stream field
1	date	streamDate
2	year	streamYear
3	month	streamMonth
4	day	streamDay

[Get fields] Enter field mapping

[Help] [OK] [Cancel] [SQL]

Output Database dw_legendvehicle

date	year	month	day
2003-01-01 00:00:00	2003	1	1
2003-01-02 00:00:00	2003	1	2
2003-01-03 00:00:00	2003	1	3
2003-01-04 00:00:00	2003	1	4
2003-01-05 00:00:00	2003	1	5
2003-01-06 00:00:00	2003	1	6
2003-01-07 00:00:00	2003	1	7
2003-01-08 00:00:00	2003	1	8
2003-01-09 00:00:00	2003	1	9
2003-01-10 00:00:00	2003	1	10
2003-01-11 00:00:00	2003	1	11
2003-01-12 00:00:00	2003	1	12
2003-01-13 00:00:00	2003	1	13
2003-01-14 00:00:00	2003	1	14
2003-01-15 00:00:00	2003	1	15
2003-01-16 00:00:00	2003	1	16
2003-01-17 00:00:00	2003	1	17
2003-01-18 00:00:00	2003	1	18
2003-01-19 00:00:00	2003	1	19
2003-01-20 00:00:00	2003	1	20
2003-01-21 00:00:00	2003	1	21
2003-01-22 00:00:00	2003	1	22
2003-01-23 00:00:00	2003	1	23
2003-01-24 00:00:00	2003	1	24
2003-01-25 00:00:00	2003	1	25

TUGAS 1

Buka preview tab pada execution result area di setiap proses object. amati input dan output data yang ada. bandingkan di setiap prosesnya. jelaskan perbedaan di setiap prosesnya.

Proses Objek	SS data input	SS data output	Keterangan
Generate rows	Tidak ada (initial step)	Kolom CurrentDate berisi tanggal "01-01-2003" diulang beberapa baris	Langkah awal menciptakan data dummy berupa tanggal CurrentDate. Data ini akan digunakan untuk langkah selanjutnya.
Add Sequences	Kolom CurrentDate	Kolom CurrentDate dan kolom tambahan incrementDay yang berisi nilai urutan mulai dari 0, 1, 2, dst.	Proses menambahkan urutan numerik incrementDay sesuai baris data dengan menggunakan sequence generator.
Calculator	Kolom CurrentDate, incrementDay	Kolom streamDate, streamYear, streamMonth, dan streamDay ditambahkan berdasarkan pengolahan nilai CurrentDate dan incrementDay.	Menghitung nilai baru seperti tanggal, tahun, bulan, dan hari dengan fungsi perhitungan yang mengacu pada data input.
Select Values	Kolom streamDate, streamYear, streamMonth, dan streamDay	Kolom streamDate, streamYear, streamMonth, dan streamDay (tanpa kolom lain)	Memilih kolom tertentu dan menghapus kolom yang tidak diperlukan (CurrentDate, incrementDay dihapus).
Database lookup	Kolom streamDate, streamYear, streamMonth, streamDay	Kolom date, year, month, dan day ditambahkan dengan hasil lookup ke database.	Proses pencocokan (lookup) ke database untuk menambahkan informasi tambahan yang relevan dengan data yang ada.
Filter rows	Kolom streamDate, streamYear, streamMonth, streamDay, date, year, month, dan day	Data terfilter (hanya satu baris tersisa atau kosong).	Menyaring data berdasarkan kriteria tertentu yang ditentukan dalam langkah filter.
Table Output	Kolom hasil dari langkah sebelumnya	Data disimpan ke dalam tabel output namun hanya baris yang lulus filter (satu atau kosong).	Menyimpan hasil akhir ke dalam tabel output pada database, menunjukkan hasil akhir berdasarkan semua transformasi sebelumnya.

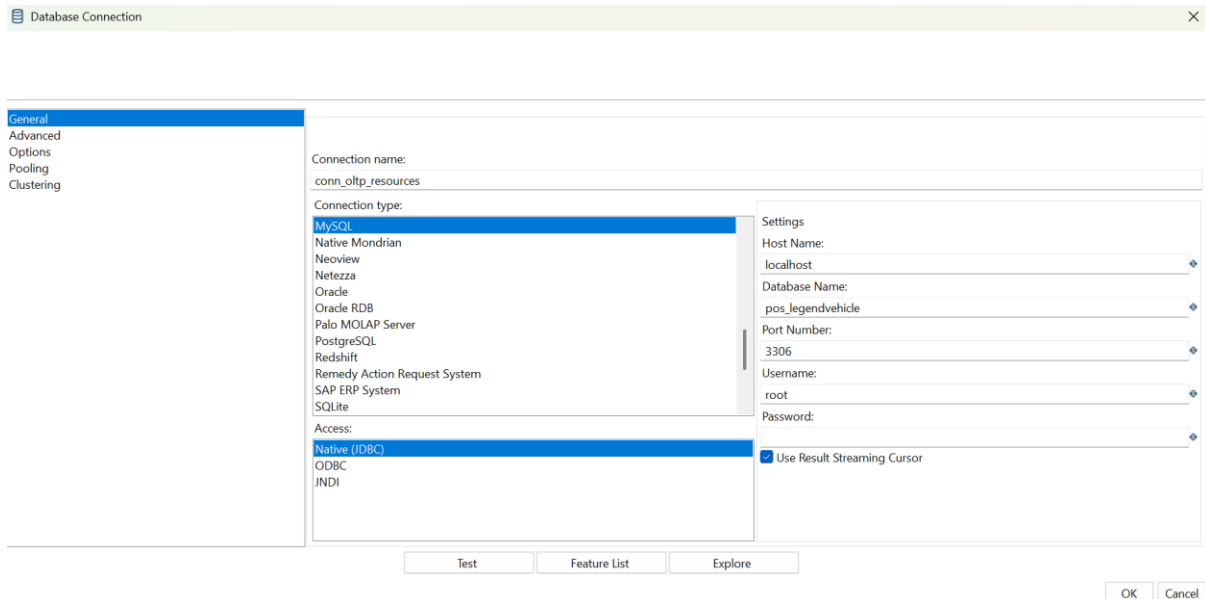
B. Dimensi Pegawai

1. Buat dimPegawai pada dw_legendVehicle

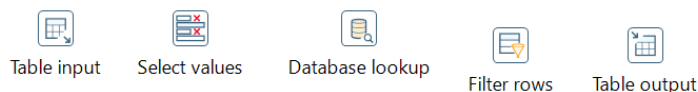
```
CREATE TABLE dimEmployees (  
  id_dimEmployees INT NOT NULL AUTO_INCREMENT PRIMARY KEY,  
  employeeNumber INT(11),  
  firstName VARCHAR(50),  
  lastName VARCHAR(50),  
  jobTitle VARCHAR(50),  
  boss_firstName VARCHAR(50),  
  boss_lastName VARCHAR(50),  
  updated DATETIME DEFAULT CURRENT_TIMESTAMP  
);
```



2. Connection Database



3. Drag and Drop Tabel input, Select Values, Database lookup, Filter rows, table output



4. Table input menghubungkan Connection pada koneksi conn_oltp_resources.

```
select * from dimemployees e  
left join dimemployees r  
on e.reportsTo=r.employeeNumber;
```

Table input

Step name: Table input

Connection: conn_oltp_resources

SQL

```
select * from dimemployees e
left join dimemployees r
on e.reportsTo=r.employeeNumber;
```

Line 3 Column 33

Store column info in step meta ☐

Enable lazy conversion ☐

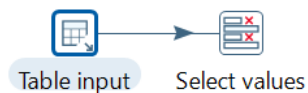
Replace variables in script? ☐

Insert data from step

Execute for each row? ☐

Limit size: 0

Help OK Preview Cancel



5. Select values mengambil data dari field employeeNumber, lastname, firstname, jobtitle, lastname_1 dan firstname_1 sebagai data stream yang digunakan pada proses ETL pada tab select & alter.

Select values

Step name: Select values

Select & Alter Remove Meta-data

Fields :

#	Fieldname	Rename to	Length	Precision
1	employeeNumber	StreamEmployeeNumber		
2	lastName	StreamLastname		
3	firstName	StreamFirstname		
4	extension			
5	email			
6	officeCode			
7	reportsTo			
8	jobTitle	StreamJobtitle		
9	employeeNumber_1			
1.	lastName_1	StreamLastnameBoss		
1.	firstName_1	StreamFirstnameBoss		
1.	extension_1			
1.	email_1			
1.	officeCode_1			
1.	reportsTo_1			
1.	jobTitle_1			

Include unspecified fields, ordered by name ☐

Help OK Cancel

Select & Alter Remove Meta-data

Fields to remove :

#	Fieldname
1	extension
2	email
3	officeCode
4	reportsTo
5	employeeNumber_1
6	extension_1
7	email_1
8	officeCode_1
9	reportsTo_1
1.	jobTitle_1

Get fields to remove

Help OK Cancel



6. Database lookup

Database lookup

Step name: Database lookup

Connection: conn_dw_destination Edit... New... Wizard...

Lookup schema: dw_legendvehicle Browse...

Lookup table: dimemployees Browse...

Enable cache? ☐

Cache size in rows (0=cache): 0

Load all data from table ☐

The key(s) to look up the value(s):

#	Table field	Comparator	Field1	Field2
1	employeeNumber	=	StreamEmployeeNumber	
2	lastName	=	StreamLastname	
3	firstName	=	StreamFirstname	
4	jobTitle	=	StreamJobtitle	

Values to return from the lookup table:

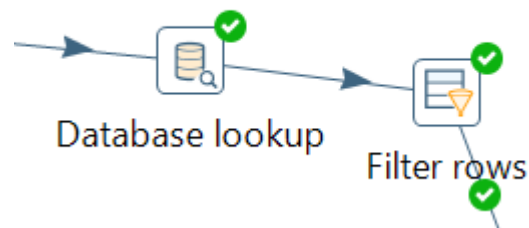
#	Field	New name	Default	Type
1	employeeNumber			Integer
2	firstName			String
3	lastName			String
4	jobTitle			String

Do not pass the row if the lookup fails ☐

Fail on multiple results? ☐

Order by:

Help OK Cancel Get Fields Get lookup fields



7. Filter rows

Filter rows

Step name: Filter rows

Send 'true' data to step: Table output

Send 'false' data to step:

The condition:

☐ employeeNumber IS NULL

AND

☐ firstName IS NULL

AND

☐ lastName IS NULL

AND

☐ jobTitle IS NULL

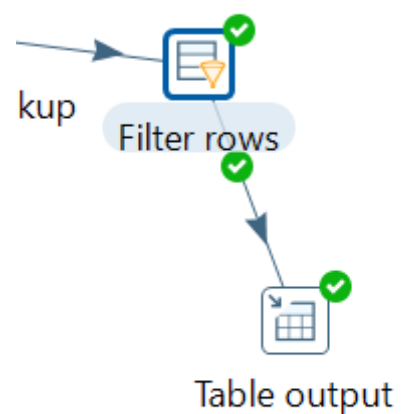
AND

☐ boss_firstName IS NULL

AND

☐ boss_lastName IS NULL

Help OK Cancel



8. table output

Table output

Step name: **Table output**

Connection: **conn_dw_destination** [Edit... New... Wizard...]

Target schema: **dw_legendvehicle** [Browse...]

Target table: **dimemployees** [Browse...]

Commit size: **1000**

Truncate table: ☐

Ignore insert errors: ☐

Specify database fields: ☒

Main options | Database fields

Fields to insert:

#	Table field	Stream field
1	employee...	StreamEmpl...
2	firstName	StreamFirstn...
3	lastName	StreamLastn...
4	jobTitle	StreamJobti...
5	boss_first...	StreamFirstn...
6	boss_last...	StreamLastn...

[Get fields]

Enter field mapping

[Help] [OK] [Cancel] [SQL]

Hasil di dalam database dw_legendvehicle pada table dimemployees

phpMyAdmin

Server: localhost:3306 » Database: dw_legendvehicle » Table: dimemployees

Extra options

	id_dimEmployees	employeeNumber	firstName	lastName	jobTitle	boss_firstName	boss_lastName	updated
☐ Edit ☐ Copy ☐ Delete	1	1002	Diane	Murphy	President	NULL	NULL	2025-03-21 13:39:18
☐ Edit ☐ Copy ☐ Delete	2	1056	Mary	Patterson	VP Sales	Diane	Murphy	2025-03-21 13:39:18
☐ Edit ☐ Copy ☐ Delete	3	1076	Jeff	Firrelli	VP Marketing	Diane	Murphy	2025-03-21 13:39:18
☐ Edit ☐ Copy ☐ Delete	4	1088	William	Patterson	Sales Manager (APAC)	Mary	Patterson	2025-03-21 13:39:18
☐ Edit ☐ Copy ☐ Delete	5	1102	Gerard	Bondur	Sale Manager (EMEA)	Mary	Patterson	2025-03-21 13:39:18
☐ Edit ☐ Copy ☐ Delete	6	1143	Anthony	Bow	Sales Manager (NA)	Mary	Patterson	2025-03-21 13:39:18
☐ Edit ☐ Copy ☐ Delete	7	1165	Leslie	Jennings	Sales Rep	Anthony	Bow	2025-03-21 13:39:18
☐ Edit ☐ Copy ☐ Delete	8	1166	Leslie	Thompson	Sales Rep	Anthony	Bow	2025-03-21 13:39:18
☐ Edit ☐ Copy ☐ Delete	9	1188	Julie	Firrelli	Sales Rep	Anthony	Bow	2025-03-21 13:39:18
☐ Edit ☐ Copy ☐ Delete	10	1216	Steve	Patterson	Sales Rep	Anthony	Bow	2025-03-21 13:39:18
☐ Edit ☐ Copy ☐ Delete	11	1286	Foon Yue	Tseng	Sales Rep	Anthony	Bow	2025-03-21 13:39:18
☐ Edit ☐ Copy ☐ Delete	12	1323	George	Vanauf	Sales Rep	Anthony	Bow	2025-03-21 13:39:18
☐ Edit ☐ Copy ☐ Delete	13	1337	Loui	Bondur	Sales Rep	Gerard	Bondur	2025-03-21 13:39:18
☐ Edit ☐ Copy ☐ Delete	14	1370	Gerard	Hernandez	Sales Rep	Gerard	Bondur	2025-03-21 13:39:18
☐ Edit ☐ Copy ☐ Delete	15	1401	Pamela	Castillo	Sales Rep	Gerard	Bondur	2025-03-21 13:39:18
☐ Edit ☐ Copy ☐ Delete	16	1501	Larry	Bott	Sales Rep	Gerard	Bondur	2025-03-21 13:39:18
☐ Edit ☐ Copy ☐ Delete	17	1504	Barry	Jones	Sales Rep	Gerard	Bondur	2025-03-21 13:39:18
☐ Edit ☐ Copy ☐ Delete	18	1611	Andy	Foxter	Sales Rep	William	Patterson	2025-03-21 13:39:18
☐ Edit ☐ Copy ☐ Delete	19	1612	Peter	Marsh	Sales Rep	William	Patterson	2025-03-21 13:39:18
☐ Edit ☐ Copy ☐ Delete	20	1619	Tom	King	Sales Rep	William	Patterson	2025-03-21 13:39:18
☐ Edit ☐ Copy ☐ Delete	21	1621	Mami	Nishi	Sales Rep	Mary	Patterson	2025-03-21 13:39:18
☐ Edit ☐ Copy ☐ Delete	22	1625	Yoshimi	Kato	Sales Rep	Mami	Nishi	2025-03-21 13:39:18
☐ Edit ☐ Copy ☐ Delete	23	1702	Martin	Gerard	Sales Rep	Gerard	Bondur	2025-03-21 13:39:18

Check all With selected: Edit Copy Delete Export

TUGAS 2

1. Buka preview tab pada execution result area di setiap proses object. amati input dan output data yang ada. bandingkan di setiap prosesnya. jelaskan perbedaan di setiap prosesnya.

Proses Object	SS data input	SS data output	Keterangan
Table Input	Data karyawan seperti employeeNumber, lastName, firstName, dll.	Data yang sama dari tabel input, tanpa perubahan.	Data awal yang akan digunakan dalam proses selanjutnya.
Select Values	Data karyawan yang sudah ada di employeeNumber, lastName, firstName, dll.	Data karyawan yang terpilih sesuai dengan kolom yang telah diseleksi.	Menyeleksi kolom-kolom tertentu dari data input untuk digunakan dalam tahap berikutnya.
Database Lookup	Data employeeNumber dan informasi lainnya.	Data tambahan yang diambil dari database, misalnya data atasan.	Pencocokan data dengan database untuk melengkapi informasi yang diperlukan, seperti bossName.
Filter Rows	Semua data karyawan yang masuk.	Data yang sudah difilter berdasarkan kondisi tertentu.	Memfilter baris-baris data yang tidak sesuai dengan kriteria yang ditentukan.
Table Output	Data yang sudah diproses, setelah tahap filter dan lookup.	Data yang sudah final dan siap untuk output ke tabel akhir.	Menyimpan hasil akhir ke tabel output atau data yang telah diproses untuk tujuan lainnya.

2. Jika proses itu di ulangi (di run kembali) apakah data akan redudant?
Hasil run tidak redundan karena proses telah diatur untuk memastikan bahwa tidak ada data yang disalin atau dimasukkan lebih dari satu kali.

- Tambahkan nama anda pada table employee di OLTP. jalankan kembali transformasi ini. Amati hasilnya, apa yang terjadi?

Server: localhost:3306 » Database: pos_legendvehicle » Table: employees

Browse Structure SQL Search Insert Export Import Privileges Operations Triggers									
		employeeNumber	lastName	firstName	extension	email	officeCode	reportsTo	jobTitle
☐	Edit		1056	Patterson	Mary	x4611	mpatterso@classicmodelcars.com	1	1002 VP Sales
☐	Edit		1076	Firrelli	Jeff	x9273	jfirrelli@classicmodelcars.com	1	1002 VP Marketing
☐	Edit		1088	Patterson	William	x4871	wpatterson@classicmodelcars.com	6	1056 Sales Manager (APAC)
☐	Edit		1102	Bondur	Gerard	x5408	gbondur@classicmodelcars.com	4	1056 Sale Manager (EMEA)
☐	Edit		1143	Bow	Anthony	x5428	abow@classicmodelcars.com	1	1056 Sales Manager (NA)
☐	Edit		1165	Jennings	Leslie	x3291	ljennings@classicmodelcars.com	1	1143 Sales Rep
☐	Edit		1166	Thompson	Leslie	x4065	lthompson@classicmodelcars.com	1	1143 Sales Rep
☐	Edit		1188	Firrelli	Julie	x2173	jfirrelli@classicmodelcars.com	2	1143 Sales Rep
☐	Edit		1216	Patterson	Steve	x4334	spatterson@classicmodelcars.com	2	1143 Sales Rep
☐	Edit		1286	Tseng	Foon Yue	x2248	ftseng@classicmodelcars.com	3	1143 Sales Rep
☐	Edit		1323	Vanauf	George	x4102	gvanauf@classicmodelcars.com	3	1143 Sales Rep
☐	Edit		1337	Bondur	Loui	x6493	lbondur@classicmodelcars.com	4	1102 Sales Rep
☐	Edit		1370	Hernandez	Gerard	x2028	ghernande@classicmodelcars.com	4	1102 Sales Rep
☐	Edit		1401	Castillo	Pamela	x2759	pcastillo@classicmodelcars.com	4	1102 Sales Rep
☐	Edit		1501	Bott	Larry	x2311	lbott@classicmodelcars.com	7	1102 Sales Rep
☐	Edit		1504	Jones	Barry	x102	bjones@classicmodelcars.com	7	1102 Sales Rep
☐	Edit		1611	Fixter	Andy	x101	afixter@classicmodelcars.com	6	1088 Sales Rep
☐	Edit		1612	Marsh	Peter	x102	pmarsh@classicmodelcars.com	6	1088 Sales Rep
☐	Edit		1619	King	Tom	x103	tking@classicmodelcars.com	6	1088 Sales Rep
☐	Edit		1621	Nishi	Mami	x101	mnishi@classicmodelcars.com	5	1056 Sales Rep
☐	Edit		1625	Kato	Yoshimi	x102	ykato@classicmodelcars.com	5	1621 Sales Rep
☐	Edit		1702	Gerard	Martin	x2312	mgerard@classicmodelcars.com	4	1102 Sales Rep
☐	Edit		9999	apip	apip	x102	apip123@example.com	1	NULL Developer

Setelah saya menambahkan nama saya pada tabel employee di OLTP dan menjalankan kembali transformasi, nama saya tidak muncul di tabel dimEmployees karena kemungkinan ada beberapa faktor yang memengaruhi, seperti proses **filter rows** atau **select values** yang tidak memasukkan data baru saya, atau **Database Lookup** yang tidak menemukan kecocokan untuk data saya. Saya perlu memeriksa lebih lanjut pengaturan dalam transformasi untuk memastikan data baru saya dapat diproses dan dimasukkan dalam output